

ATPL EASA STUDY GUIDE | 021 - AIRFRAME & SYSTEMS

Chapter 22: FADEC, Starting Systems & Parameters (N1, EGT, EPR)

Syllabus Source: CAE Oxford (pp. 257-310 (Pwr)) | Pass Target: >= 90% AviationExam

1. FULL AUTHORITY DIGITAL ENGINE CONTROL (FADEC)

FADEC has complete, unrestricted digital electronic control over all engine fuel metering, geometry, and operational functions with NO MECHANICAL BACKUP:

- Dual-Channel Redundancy: Consists of an Electronic Engine Controller (EEC) with two identical, independent digital channels (Channel A and Channel B). One channel is active while the other is in hot standby. Channels alternate control automatically on each engine start.
- Dedicated Power (PMA): FADEC is completely self-powered above ~10-15% N2 by its own engine-driven Permanent Magnet Alternator (PMA). Even if aircraft dual AC/DC electrical systems fail completely, FADEC continues operating with zero interruption!
- FADEC Control Functions: 1. Fuel metering (FMV); 2. Idle speed control (Flight idle, Ground idle, Reverse idle); 3. Transient bleed valves & VSVs; 4. Turbine active clearance control (ACC); 5. Thrust reverser deployment; 6. Engine limit protection (prevents N1/N2 over-speed and EGT over-temperature during auto-starts).

2. ENGINE STARTING SYSTEMS & ABNORMAL START DRILLS

Modern jet engines use a pneumatic starter motor powered by APU, crossbleed, or ground air cart to spin the HP spool (N2):

- Standard Start Sequence: 1. Pneumatic starter valve opens -> Starter cranks HP spool (N2 rotation); 2. At maximum motoring speed (typically 15% to 25% N2), fuel flow and ignition are switched on; 3. Light-off occurs -> EGT rises rapidly; 4. At self-sustaining idle speed (~50% N2), starter valve closes and ignition turns off.

Abnormal Start Condition Cockpit Instrument Symptoms		Immediate Flight Crew Emergency Action
Hot Start	EGT rises rapidly and threatens or exceeds maximum start limit before reaching idle speed.	ABORT START! Cut off fuel immediately; continue dry motoring engine with starter to blow cooling air through core.
Hung Start	Engine lights off and EGT rises, but RPM stabilizes well below self-sustaining idle speed (N2 hangs).	ABORT START! Cut off fuel; starter alone cannot overcome compressor drag.
Wet Start	Fuel is injected and flows into engine, but NO IGNITION occurs within specified time (EGT does not rise).	ABORT START! Cut off fuel; motor engine for at least 30-60 seconds to clear unburned fuel vapor before reattempt!

3. PRIMARY ENGINE THRUST PARAMETERS: N1, EPR & EGT

Turbine thrust cannot be measured directly in flight. Cockpit instruments indicate computed thrust indicators:

- N1 (Low Pressure Spool / Fan Speed %): PRIMARY thrust setting parameter on high-bypass turbofan engines (CFM56, GE90, LEAP). Because 80% of total thrust is produced by the cold bypass fan, N1 RPM correlates directly with aerodynamic thrust.

- Engine Pressure Ratio (EPR): PRIMARY thrust setting parameter on Rolls-Royce (Trent) and Pratt & Whitney engines. $EPR = \text{Total Pressure at Turbine Exhaust (P}_{t7}) / \text{Total Pressure at Engine Compressor Inlet (P}_{t2})$. An EPR of 1.50 means exhaust pressure is 1.5 times inlet ram pressure.
- Exhaust Gas Temperature (EGT): Measured between turbine stages or at turbine exhaust by chromel-alumel thermocouples. Indicates the thermal stress and health of the turbine. The primary life-limiting factor of gas turbine engines.

[!] EXAM TRAP (AVIATIONEXAM) - N1 vs EPR Engine Pressure Probe Icing Trap

- EPR engines rely on a heated P_{t2} pitot probe in the engine intake nose cone. - If the P_{t2} probe inlet blocks with ice in icing conditions: The probe senses artificially high pressure -> EPR indicates ERRONEOUSLY HIGH -> Pilot throttles back -> Engine delivers dangerously insufficient take-off thrust! - (This was the fatal primary cause of the Air Florida Flight 90 crash in Washington D.C.)!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): FADEC Architecture & EPR vs N1

1. Dual-Channel EEC: Active Channel and Standby Channel cross-talk continuously. If the active channel faults, the standby channel takes over seamlessly. 2. PMA Alternator: Dedicated permanent magnet alternator powers FADEC independently of all aircraft electrical buses. 3. Thrust Setting Datum: - Rolls-Royce & Pratt & Whitney: Primary thrust parameter is EPR (Pt7/Pt2). - CFM & General Electric: Primary thrust parameter is N1 fan speed.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is a 'hung start' on a jet transport engine?

- [A]: The starter motor fails to engage
- [B]: The engine ignites normally, but rotational RPM fails to accelerate up to self-sustaining idle speed, stabilizing at an intermediate sub-idle value
- [C]: EGT exceeds limits immediately
- [D]: The propeller fails to unfeather

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In a hung start, fuel lights off and EGT rises, but the engine lacks sufficient torque to accelerate past starter cut-out speed up to self-sustaining idle, hanging at low RPM.

Q2: In a Full Authority Digital Engine Control (FADEC) system, what is the dedicated source of electrical power

- [A]: The aircraft main 28V DC battery
- [B]: An engine-driven Permanent Magnet Alternator (PMA) mounted on the engine accessory gearbox
- [C]: The aircraft TRU
- [D]: The APU generator

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. FADEC incorporates its own dedicated PMA generator on the gearbox. Once above ~10-15% N2, FADEC is 100% self-powered, immune to any aircraft electrical failures.

Q3: On an aircraft that uses Engine Pressure Ratio (EPR) as its primary thrust indication, what is the mathematical formula for EPR?

- [A]: Ratio of compressor exit pressure to burner pressure
- [B]: Ratio of turbine exhaust total pressure (P_{t7}) to compressor inlet total pressure (P_{t2})
- [C]: Ratio of fan speed to core speed
- [D]: Fuel flow divided by air flow

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. EPR is defined as the total pressure at the turbine discharge / exhaust nozzle divided by the total pressure at the engine intake face (P_{t7} / P_{t2}).